PRAXIS

Task authoring requirements and handover

Version 2.0 | 10 November 2026

One common chatbot. One reusable Classic framework.
Template-driven fields, policies, review and validation.

Latest UI decision: a connected horizontal process graph below both Classic and Chatbot.

Includes field-by-field guides, user journeys, architecture, implementation gaps, memory and guardrails, policy ingestion, Expert/PM review, Vertex MaaS selection, validation and continuation decisions.

Local source inspected: de758ba. New graph is a mockup. No deployment, paid model call or training is claimed.

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Praxis task-authoring handover v2.0

Prepared 10 November 2026, America/Chicago. Product requirements, design and implementation handover. Latest local product commit inspected: de758bac9247fe7c53d36f033e225964f34f75f0 on feat/conversational-case-authoring. Source repository: https://github.com/vasudevansundarababu-dev/adi-praxis.git; owner's source branch: feat/catalog-published-defaults. No push, merge, deployment, paid model call or training is included in this documentation delivery.

Read this before building

Praxis helps qualified SMEs author realistic benchmark tasks. The SME supplies substantive case facts, evidence, recorded tool behaviour, the reference solution and grading judgement. The chatbot facilitates that work; it must not become a second answer-author. Anvil executes and evaluates exported tasks; Praxis acceptance alone does not prove executable benchmark readiness.

There is ONE shared chatbot capability for ALL templates and ONE reusable Classic authoring framework. A template contributes its versioned field definitions, requiredness, pick lists, examples, tools, policies, reference shape and validations. Do not build fourteen independent chatbots or fourteen unrelated Classic pages. Both views read/write the same canonical task. Not every current generic template has the required typed field implementation yet.

Project creation chooses Classic, Chatbot or Both. Template configuration determines Single Agent or Multi Agent before assignment. SMEs and reviewers cannot choose an agent mode. In Both, switching view never creates another task, reference answer, rubric or payable submission.

Latest layout decision

The Flow preview runs horizontally, left to right, BELOW the authoring area in BOTH Classic and Chatbot. It is a proper connected graph: agent nodes, diamond decisions, labelled branches, business human-review nodes, correction loops, limits, escalation and final output. It must not be replaced with stacked prose cards or text arrows. At narrow widths reflow the same connections into a compact accessible layout; never shrink labels until unreadable. Graph shape and task data remain identical when the editor changes.

The human-review node is a business step inside the authored case. Praxis Expert review and PM approval of the task remain outside that graph. These responsibilities must not be conflated.

Scope and delivery status

- EXISTING: project/template/JD/staffing, PM/Admin policy intake and rule publication, KYC fields/checks, autosave, one-stage authorised review/rework, conditional earnings, immutable accepted dataset and export services.

- LOCAL IMPLEMENTED: project Classic/Chatbot/Both configuration; server view enforcement; shared consent/declarations; persisted conversation/proposals and unsent answer; revision/policy checks; common submission controls; save barriers and Classic preview retention. Tested locally; not deployed here.

- DESIGN / TO BUILD: schema-driven full Classic forms for all templates, horizontal workflow graph with decisions/human interactions/loops, fully adaptive template-aware clarification beyond current deterministic prompts, SME policy submission/review bridge, configurable Expert-to-PM stage, per-template executable adapters and new managed-open-model provider adapter.

- EXCLUDED: two-SME adjudication, SME versus LLM answer competition, model training platform, autonomous operational banking actions, cloud provisioning and paid benchmark execution.

Document map

01 Requirements and acceptance; 02 System/chatbot architecture; 03 Project setup fields; 04 SME screens and section fields; 05 Shared templates and graph contract; 06 Chatbot memory and guardrails; 07 Policy intake/RAG; 08 Expert/PM review; 09 User journeys; 10 Existing-code change impact; 11 Vertex model choice and cost; 12 Build order and validation; 13 Help text and operating guides; 14 Decision register and continuation handoff; 15 KYC field contract; 16 Template catalogue supplement. Existing per-template six-section guides accompany the package as appendices.

This v2 handover resolves conflicting historical notes. In particular: remove unrestricted SME agent-mode pickers; remove blanket chat-default assumptions; never put platform approval into the investigation graph; keep human substantive authorship; put the connected graph below BOTH editors; use one template-driven authoring system. Older mockups are historical references.

01 Requirements and acceptance

Purpose and actors

Admin manages system configuration and authorised final actions. PM prepares projects, policies, JD, staffing, targets and exports. SME authors assigned cases. Expert evaluates domain correctness and the SME's added policy material, may make audited corrections or return work. PM provides final approval in the target two-stage flow. One authorised person may hold Expert and PM responsibilities, but the decision records remain distinct. Do not grant PM new permissions simply by hiding a button.

Demo flow: SME -> authorised PM/Admin final review. Target alternatives: SME -> PM; SME -> Expert -> PM. Two independent authors and SME/LLM comparison are explicitly excluded from MVP.

Functional requirements

ID	Requirement	Acceptance evidence
PR-01	Project authoring experience is required for new AUTHORING projects: CLASSIC, CHATBOT or BOTH.	UI/API refuse omission; persisted selection survives reload. Legacy NULL is preserved.
PR-02	Published template fixes agent mode and schema version before assignment.	No SME/reviewer mode picker; server refuses unsupported workflow/export.
PR-03	Shared template registry powers every Classic screen and the common chatbot.	A new template is registered through schema/configuration; no copied chatbot or page.
PR-04	Both editors share task ID, output, reference, rubric, policy snapshot and revision.	Change in one view appears in the other after a verified save; no duplicate task.
PR-05	Preserve typed answers during section, view and agent-preview transitions.	Final keystroke immediately followed by navigation is saved; failure retains editor and text.
PR-06	Chat memory includes current fields, relevant prior answers, unresolved questions and actual rework feedback.	Missing SSN/Tax ID in a later answer creates contextual clarification without forgetting earlier evidence.
PR-07	SME creates the substantive answer; chatbot extracts, asks and checks.	Requests to generate the answer or answer unrelated questions do not produce substantive content.
PR-08	Confirm proposed coordinated changes before mutating case output.	Before/after and affected fields visible; stale task/policy proposals cannot apply.
PR-09	Horizontal connected graph below both editors represents the same saved task.	Parallel joins, decisions, human steps and bounded loops have visible connected routes and field bindings.
PR-10	Policy is project-level, immutable by version and published only through authorised review.	Pending or random uploaded documents cannot become authoritative context.
PR-11	SME may propose bank/customer policy additions for review.	A permission-scoped submission queue exists; approved rule versions, not raw upload labels, become usable.
PR-12	View as Agent is a server projection.	Reference/rubric/private discussion and future/other-role artifacts are absent, including downloadable payloads.
PR-13	Eight-section target experience is common across templates with domain-specific controls.	Every applicable content section has editable fields or a clear Add action; no first-tab-only editor.
PR-14	Expert/PM review is tied to a submitted immutable revision.	Changes invalidate affected review/approval; the reviewer never approves an unspecified latest draft.
PR-15	Rework returns the same task, preserving comments and provenance.	No second accepted count or earning for resubmitting the same accepted unit.
PR-16	Payment uses existing eligibility and ledger controls.	Approval may create one eligible earning; actual payout status remains separate.

ID	Requirement	Acceptance evidence
PR-17	Target count/date controls intake, closure and export.	Only approved, eligible immutable records included; incomplete/in-review cases stay visible and excluded.
PR-18	Anvil contract validation is separate from dataset export.	Missing adapter, unsupported graph or invalid binding blocks executable export with a precise reason.

States and review responsibilities

User-facing task states are Authoring, Review/Approval Pending and Approved. Within Pending store a responsible stage (Expert review or PM approval), an assigned reviewer and the submission version. Rework returns to Authoring with comments. A final declined/withdrawn administrative record may be necessary internally, but do not invent an additional user-facing state without a decision. Existing status enums and review routes are not already equivalent to this target model; migrate deliberately.

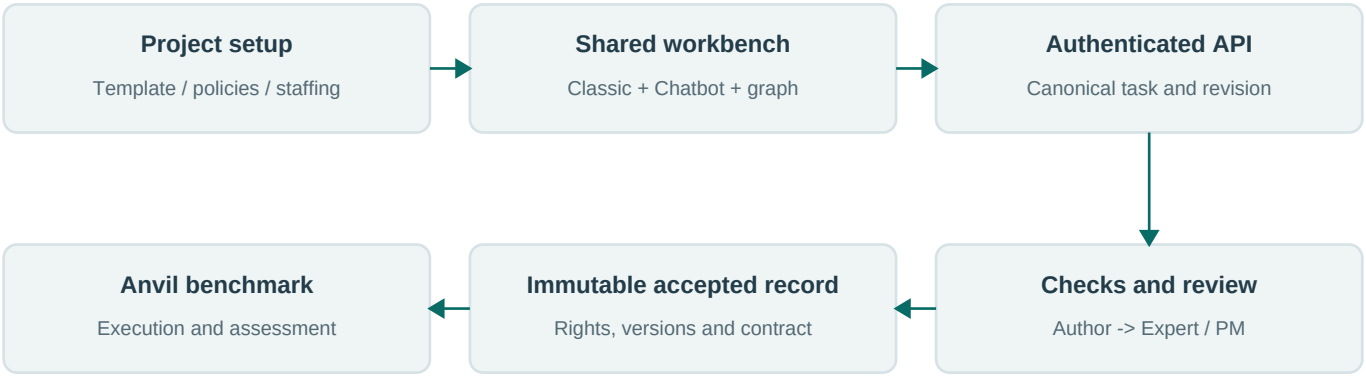
Template single/multi mode, project authoring experience and project review route are three independent settings. A multi-agent template can be authored in Classic or Chatbot. A Classic task can still contain operational human handoffs.

Nonfunctional requirements

Proposed pilot targets: p95 first acknowledgement under two seconds for accepted messages, p95 completed small clarification under 15 seconds, 99.5% monthly authoring availability, and accessible labelled controls at 1024/736/390/320 widths. These are targets, not measurements or purchased SLAs. Persist the user message before dispatch; display generation pending/failure honestly. Account for reasoning tokens, retries and rejected responses. Project/tenant ownership applies to files, turns, retrieval and exports. Publish retention, consent and document-rights notices before collection. Never log credentials or whole sensitive uploads in model telemetry.

A hosted open model is not free usage. No model recommendation authorises enabling a service or spending money. Browser typing restrictions cannot prove that an SME did not use another LLM. A valid JSON response or citation does not prove semantic correctness.

02 System and chatbot architecture



Platform boundaries

Praxis owns project setup, staffing, authoring, policy review, task acceptance, payable eligibility and immutable accepted records. Anvil owns benchmark execution, environment qualification, assessment and benchmark receipts. The authoring chatbot is a facilitator in Praxis; it is not the benchmark agent or a task approver.

Application flow: PM/Admin project setup -> assigned SME -> shared authoring workspace -> authenticated Praxis API -> canonical task/revision service -> check/submit -> Expert/PM decision -> immutable dataset -> contract-qualified export -> Anvil. Existing payment services consume eligible acceptance events, not chatbot responses.

Chat flow: typed answer -> ownership/consent/scope checks -> persist turn -> durable job -> context assembly -> model adapter -> schema/source/visibility checks -> pending proposal -> SME confirmation -> atomic task revision -> map/rubric/readiness refresh. Provider work happens outside the database transaction. An uncertain request must not be paid-retried automatically.

Reusable modules

Module	Responsibility	Boundary
Template registry	Versioned fields, enums, requiredness, tools, examples, output/rubric/graph contracts and adapter capabilities.	Drives both editors; no template-specific chatbot forks.
Shared draft service	Optimistic revisions, complete canonical output, autosave, field provenance and edit invalidation.	One task record; chat draft text is not confirmed output.
Classic renderer	Renders configured controls, repeated records, help and validation.	Optional domain plug-ins operate through the same draft service.
Chat facilitator	Requests missing SME content and proposes placement of supplied words.	No case-answer generation, policy publication or arbitrary tool use.
Context builder	Joins approved policy, current case, relevant turns and unresolved dependencies.	Enforces tenant/project/task/visibility before retrieval or model input.
Grounding/guard layer	Validates schema, references, source spans, dependencies and deterministic rules.	Model confidence is never an approval signal.
Policy service	Document versions, extraction, candidate rules, expert review and published sets.	An uploaded document is untrusted until approved.
Projection service	Builds full-SME, Expert/PM and per-role agent views.	Private rubric and hidden answers never enter an agent package.
Review engine	Immutable submission decisions, rework, role authority, approval invalidation.	Business human nodes are separate from platform review stages.
Export service	Dataset packages, contract adapters, compatibility and blocking reasons.	A graph visualisation is not executable qualification.

Persistent memory design

Store raw author turns, confirmed structured fields, current field provenance, unresolved issues, rule-version links, pending proposals, review comments and declared revision. Scope all memory to tenant, project and task; do not mix cases because the same SME works on both. A user deleting or correcting a fact creates an audited supersession, not silent retention as current truth.

Current code uses TaskInterviewTurn, TaskDraftRun and TaskInstance output/drafting metadata, with bounded direct context. The current implementation can fail at the history/context limit; it does not provide an unlimited ChatGPT-style memory guarantee. A reviewed summary store and permission-filtered RAG are proposed extensions. Raw history remains the source of truth; summaries must identify covered turn IDs, source spans, outstanding contradictions, version and superseded facts.

Dependency reconciliation

An SME adding a missing identifier affects the scenario, evidence-gap inventory, relevant tool responses, reference findings, applicable rule links and rubric criteria. Compute an affected-field graph from the template; show proposed changes and unresolved dependencies. Do not silently fabricate a tool response, expected conclusion or grading weight. When the SME changes a fact, invalidate the prior reconciliation/declaration and any affected expert decision. Auto-updating a map from confirmed structured steps is safe; inventing new substantive steps is not.

Provider abstraction

Use an internal ConversationAssistant interface, not an LLM-specific page. Input contains a versioned task/context snapshot and allowed field contract. Output contains proposed changes, source links, clarification targets and unresolved issues; only allowed sections are writable. A new Vertex MaaS adapter is a replacement behind this interface, with provider-specific schema translation, token accounting, timeout/error classification and contract tests.

Keep the existing Vertex Claude adapter for compatibility until a reviewed migration. Current tests use controlled/mock providers. Do not claim that adaptive live-model quality, native multi-agent execution, large-library RAG or new MaaS access is validated by those tests.

03 Project creation: field-by-field specification

PM/Admin prepares the project before any SME authors tasks. Reuse current project, template, JD, policy and staffing services. Add only missing configuration/permissions and renderers.

Field/control	Meaning and what the user does	Validation / help
Project name	Give the assignment a clear business name.	Required. Example: Customer onboarding benchmark, November set.
Domain and published template	Choose the task family and exact published version.	Required; unpublishable/draft versions cannot silently become assignments.
Agent type	Read the template-defined Single or Multi value.	Display-only here after selecting the template; never repeated as an SME mode choice.
Authoring experience	Choose Classic, Chatbot or Both.	Required for new authoring projects. Both uses one task, not two authoring submissions.
Review route	Choose SME -> PM or SME -> Expert -> PM in target design.	New workflow configuration; current one-stage permissions are not sufficient evidence of implementation. Demo uses authorised Admin final acceptance.
Task target	Specify accepted tasks required, not raw submissions.	Positive validated count. Rework does not increase it.
Coverage dimensions and quotas	Specify customer types, risks, locales, edge cases and required coverage.	Sum/target rules and available slots must reconcile. A reached count with missing coverage is not necessarily complete.
Maximum per SME	Limit assigned workload and concentration.	Must fit project target and staffing policy; respect existing assignment checks.
Start/end dates	Define assignment window and project deadline.	End after start; timezone shown. Agree rework grace separately; never imply that existing task autosave expires at the end date.
Time target	Set expected effort per task.	Planning/pay input, not an automatic timeout on the author.
Project policies	Add authoritative base documents and review their extracted rules.	Project scope; only published versions are active. Show gaps before project readiness.
Bank/customer category	Label the document's stated authority and scope.	A label does not establish authenticity. A customer document does not override bank policy by default.
SME job description	Attach an existing JD or prepare one from template skills.	Existing JD CRUD can be reused. Automated JD generation is proposed, not proven by current code. PM reviews generated wording if added.
Required skills/jurisdiction/language	Define qualifications used by matching.	Reuse eligibility and staffing gates. Distinguish hard constraints from preferences.

Field/control	Meaning and what the user does	Validation / help
Rate/currency and payment terms	Set authorised compensation terms.	Reuse approved rate cards, pay routes and eligibility. Do not promise a payout merely because a chatbot is complete.
Matched SME list	Inspect qualified candidates and assignment rationale.	PM/Admin assigns; chatbot cannot recruit, assign or bypass identity/onboarding requirements.
Expert reviewers	Assign qualified industry experts when route requires them.	Target extension; authorised access only to assigned project/tasks/documents.
Final approver	Assign authorised PM/Admin responsibility.	Current API role restrictions must be explicitly extended and tested before PM approval is promised.
Create/save/submit project	Review configuration, then create and staff it.	Required fields, policy readiness and staffing constraints shown with actionable errors.

Project configuration lifecycle

Persist authoringExperience on the Project; the local implementation uses nullable CLASSIC/CHATBOT/BOTH. Legacy NULL retains previous Both access without silently enrolling existing tasks into new declarations. Ordinary legacy edits must not force a migration. Mode changes after task assignment are blocked in the local slice; an active-project migration needs a separate preview, authority, compatibility and rollback plan.

The template should declare a versioned authoring UI profile and export capability. Do not mutate a published template's shape under active tasks. Project creation pins template version, instruction version, review route and policy baseline. Every submission records what was actually used.

PM help example

"Choose how SMEs will author. Classic provides guided fields. Chatbot asks questions and organises the SME's own words. Both allows switching on the same task. This setting does not choose how many agents execute the benchmark."

04 SME workbench: screens and fields

Common shell for every template

Show project/template/version, assignment, task status, current saved revision, section X of 8 and workflow step/role count. Only a Both project shows Classic/Chatbot tabs. Keep Instructions PDF, View as Agent, save status and validation accessible in both. Flow preview is a connected horizontal graph BELOW the authoring area. The same graph appears after changing editor; no separate graph copy.

Selecting a role/step updates its authoring context. Selecting a decision/human/loop shows the relevant routes and conditions. A business human node is an authored scenario interaction; it is not the Expert reviewing the task or the PM approving it. In Classic show appropriate controls; in Chatbot show the conversation and persistent typed-answer composer. Do not restrict editability to the first tab.

Current KYC uses six grouped sections W01-W06 and non-KYC templates use the shared generic editor. The eight-section specification below is the common target renderer, not a claim that all existing pages already implement it.

Section 1: Assignment and brief

Field	Meaning / action	Example / check
Title	Name the work the benchmark agent must do.	Resolve inconsistent applicant identity evidence. Required.
Trigger	Choose the template event that starts the work.	KYC: NEW_ONBOARDING or PERIODIC_REFRESH.
Business context	Explain the situation and assignment without giving the answer.	Identity and address records have different family names.
Customer/subject type	Identify the entity or subject category.	Individual applicant; scope determines applicable rules.
Product/channel	Describe relevant service and submission channel.	Current account / online application, if supplied.
Jurisdiction/language/as-of date	State the task's boundaries and relevant date.	Do not let the bot infer a country from an identifier label.
Difficulty/estimated steps	Set template-supported complexity and expected work size.	KYC difficulty 1-5. Not an execution receipt.
Coverage tags/edge case	Classify the case into the assignment's coverage cell.	Missing identifier and inconsistent name; use configured pick lists.

Section 2: Evidence and parties

Field	Meaning / action	Example / check
Evidence ID/type	Give every input a stable ID and document/record type.	E1 identity record; E2 address record. No duplicate IDs.
Source/issuer/date	Identify origin, issuer and date known to the case.	Missing issuer stays unknown; do not invent provenance.
Extracted fields	Record the actual relevant text/values.	Family-name values differ. Use authorised test data.
Party ID/name/kind/role	Identify who the evidence concerns.	P1 / applicant / PERSON / CUSTOMER.
Relationships/ownership	Link parties and quantities when the template needs them.	Supported ownership percent; no unsupported beneficial-owner inference.
Missing/inconsistent evidence	Explicitly describe absence or contradiction.	SSN/Tax ID absent; requirement applicability not yet established.
Case-source attachment	Upload authorised supporting source material through the source route.	An attachment is evidence, not an approved project policy.

Empty evidence collections must display an Add action and explanatory text. Adding a record opens its actual fields. Do not create decorative text areas unrelated to the contract merely to make a tab look editable.

Section 3: Environment and recorded tools

Field	Meaning / action	Example / check
Environment choice	Select supported tool environment or explicit none.	None for a pure structured decision; tool environment requires executable binding later.
Tool catalog	Choose only template-approved actions.	document_check, registry_lookup, request_info.
Response ID/tool	Identify the exact allowed recorded call.	R1 / document_check.
Arguments	Enter schema-valid inputs to that call.	Reference the correct party/document ID.
Response kind/content	Write what the environment returns.	CLEAR/HIT/NEAR_MATCH/DATA/ERROR for KYC.
Response visibility/reveal stage	Declare when and to whom the recorded response is available.	Not all tool outcomes belong in the initial agent-visible prompt. Target graph/adapter extension.
Reset/failure behaviour	Define repeatability, unavailable tools and error handling.	Missing tool outcome remains unassessed, not fabricated success.

A recorded response is not evidence that a live tool executed. The author supplies it as part of the test environment. A benchmark-specific adapter must qualify replay/isolation before real execution.

Section 4: Applicable project policies

Field	Meaning / action	Example / check
Published rule/version	Select the exact approved project rule.	PR identifier is an example only until linked to a real published version.
Source page/clause/quote	Locate the authority supporting that rule.	Keep a faithful quotation and locator.
Applicable step/role	Bind the rule to the work it governs.	Identity discrepancy assessment step.
Applicability rationale	Explain jurisdiction, category, scope and exceptions.	Why the rule applies to this applicant; do not infer a universal SSN requirement.
Conflict/exception	Record competing rules or unmet conditions.	Needs expert clarification; do not silently pick precedence.
Additional bank/customer policy	Propose a document for authorised review.	Pending until qualified/reviewed/published. No immediate authority.

Section 5: Expected work and process

Define steps, expected results, reasons, findings, evidence references, recommendations, risk rating where applicable, rationale and authorised business handoff. The SME writes the reference solution; the bot asks for missing parts. For multi-agent templates also define roles, permissions, input/output artifacts, joins, decisions, human responses and bounded returns. See chapter 05 for graph fields.

A missing fact may imply an unresolved finding; it does not justify inventing a tool result or policy obligation. Changes to evidence flag dependent solution and rubric fields for reconciliation. Keep reference/private grading out of the agent brief.

Section 6: Allowed outcomes and values

Use the template's pick lists. For KYC: RECOMMEND_APPROVE, RECOMMEND_REJECT, REQUEST_INFORMATION, ESCALATE; risk LOW/MEDIUM/HIGH. These are recommendations or routing decisions, not a real account action. If no option fits, record an unsupported selection request for review rather than changing the enum. Non-KYC workflows use their own schema; do not force a banking disposition into a marketing task.

Section 7: Rubric

Criterion ID/name identify what is assessed. Weight sets its contribution; use 100 total for KYC. Pass condition must be observable against case evidence, tool calls or output. The target renderer also needs explicit partial/zero-credit anchors, critical-failure scoring semantics, evidence/rule/step bindings, common errors and acceptable variations. These richer bindings are not all fields in current KYC types; add them through a versioned schema/adapter, not an undocumented rename.

Example: "Identify the E1/E2 family-name discrepancy and cite both records." The SME supplies its weight and critical judgement. Do not grade identical wording when a different evidence-supported answer is acceptable. Mark a critical failure separately from a zero-weight criterion; define whether it caps or fails the task.

Section 8: Check and submit

Show missing required fields, invalid references, policy applicability gaps, unsupported graph/export features, weights, privacy flags and unresolved proposals. Each finding links to the affected field/node. Save before checks and use the current revision. Show authorship/source declarations, pending policy additions, View as Agent and the intended review route. Submit sends one frozen version to review; it does not approve it or pay it.

Chatbot-specific fields and actions

Conversation history shows authored answers and safe facilitation messages. Your answer is a persistent draft; it is not automatically a case fact. Send stores the turn and queues facilitation. Proposal review shows the supplied wording, intended fields, source/turn links, affected dependencies and unresolved issues. Confirm applies atomically; discard preserves the canonical task. Cancel/timeout must not lose typed work or imply a successful response. Provide retry only with clear attempt/spend semantics. A section jump changes context, not the stored answer's ownership.

05 One template-driven platform and graph contract

Shared template architecture

Every published template has a versioned authoring profile: identifier/version, agentMode, enabled task modes, field schema, section layout, requiredness/conditional rules, pick lists, tools/input schemas, source permissions, examples/help, reference solution shape, rubric shape, deterministic check registry and export capabilities. This profile drives ONE Classic renderer and ONE chatbot context builder. A schema adapter maps the author's terminology to the existing canonical contract. Do not generate new field names from chat prose.

Use shared primitives for text, long answer, number, date, pick list, multi-select, evidence record, repeated array, policy citation, tool call, criterion and graph binding. A domain plug-in is permitted for a specialised control, but it uses the same save/consent/check/projection APIs. Template registration tests prove that both editors expose required fields and preserve unknown legacy metadata. A template with an unsupported field type is disabled with a diagnostic; it must not fall back to an apparently complete blank form.

Single-agent templates use steps and dependencies for one executor. Multi-agent templates additionally declare roles, handoff artifacts, tool permissions, joins and branch/loop contracts. One hundred independent cases are one hundred trials; record count does not itself make a task multi-agent. Authoring experience, agent mode and human review route remain independent.

Horizontal graph layout

At desktop widths the graph runs left to right beneath the authoring area: case evidence -> parallel specialists -> evidence decision -> consolidation -> human-referral decision -> business review -> final outcome. Missing evidence branches downward to a request and returns to the case input. Corrections return above the main path to consolidation. Exhausted or unresolved cases exit to a case lead. Use decision diamonds, role rectangles, a distinguishable human node, labelled directional edges and terminal boundaries. The same selected role and confirmed wording are shared by Classic and Chatbot.

Keep the graph shorter than the previous tall vertical design; the current desktop reference is roughly 620 pixels tall. That is a design dimension, not a hard limit for all workflows. On smaller screens reflow connections into a readable compact layout or accessible graph list. Do not scale fixed text into illegibility, crop loops or force the whole page to scroll horizontally. Large enterprise graphs may need a separately designed pan/zoom workspace.

Proposed WorkflowGraph v1 fields

This is a TARGET extension. Current KYC KycStep only supports TOOL/REASONING and dependsOn; it cannot encode the full graph below. Preserve current data until a versioned migration and export adapter exist.

Field	Author/reviewer meaning	Validation
schemaVersion / templateVersion	Identify the graph and field contract.	Pin both; refuse unsupported versions.
nodes[].id/type/label	Stable node, kind and visible name.	Unique ID; allowed kinds: input, agent_step, decision, human_step, join, output, escalation.
nodes[].roleId	Executor responsible for an agent step.	Must resolve to declared role; no autonomous human impersonation.
instruction / inputRefs / outputArtifact	Specify work, available evidence and required handoff.	Inputs known and visible; output schema explicit.

Field	Author/reviewer meaning	Validation
policyRuleVersionIds / rubricCriterionIds	Bind authority and assessment.	Resolve to approved applicable rules and current rubric.
decision.conditions[]	Define branches and conditions.	Each edge has an observable condition, explicit else/unknown route and approved basis.
join.requiredInputs	Define which outputs are required before continuation.	No ambiguous "wait for all" when some branches are optional.
human.actorRole / responseSchema	Define business authority and human interaction.	Expected choices/timeouts/recorded response or supported human-in-loop runtime contract required.
edges[].from/to/port/label	Define a connected directed handoff.	Both endpoints exist; compatible artifact schemas; all decisions labelled.
loop.id/returnEdgeId/maxIterations	Identify a bounded correction route.	Positive configured limit; no unlimited cycle. Two is illustrative, not a universal rule.
loop.exitCondition/exhaustedTarget	Define success and unresolved behaviour.	Both routes terminate or escalate; no silent successful outcome.
visibility / revealCondition	Limit artifacts per role/stage.	Private rubric/solutions excluded; future evidence revealed only under the contract.

Do not encode a loop as a cyclic dependsOn list in the existing DAG model. Store transitions and loops separately and validate bounded reachability. A graph node for "human reviewer" without an interaction contract is authoring design only, not an executable benchmark feature.

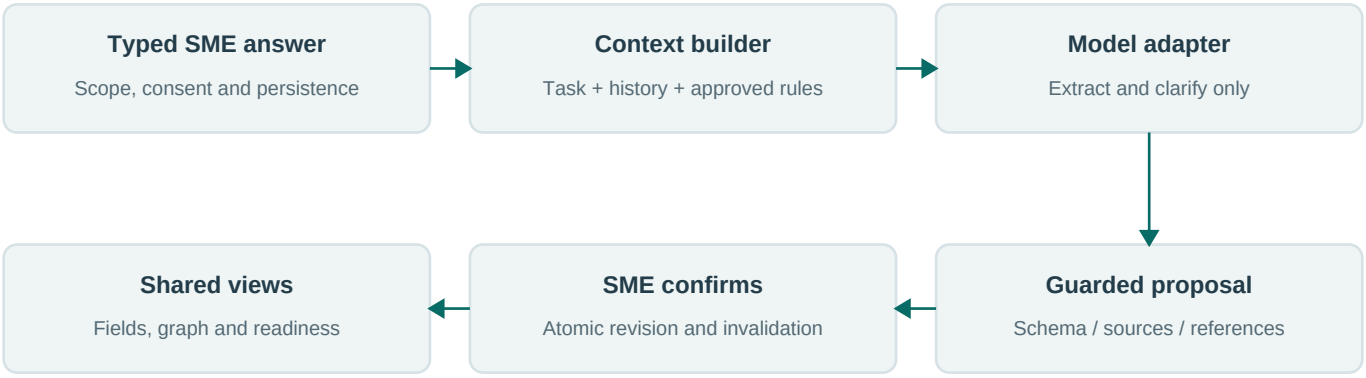
Graph checks

Validate reachability from input, at least one valid terminal output, stable references, role permissions, required joins, branch coverage including unknown/error, bounded cycles and compatible handoff schemas. Flag a specialist whose result is ignored, a decision with no else/unknown path, a loop without an exhausted exit, and private-data leakage through a handoff. Review whether an authorised human step is actually required by the case; never insert platform review into the business graph.

Anvil mapping

Evidence -> source artifacts and visibility. Constraints -> permitted actions and policy requirements. Tools/responses -> environment call contract and recorded fixtures. Agent steps -> actions/stages with input/output bindings. Reference solution -> private reference_output. Rubric -> criterion/evidence/rule bindings with defined scoring. Graph branches/loops/human interactions -> supported versioned execution contract. Unsupported elements must block executable export. Do not flatten decisions or human judgement into a single prose stage and claim equivalent benchmarking.

06 Chatbot, memory, hallucination and adversarial controls



Authoring policy

The assistant may clarify the current field, quote approved rules with sources, identify missing information, organise the SME's own text and propose source-linked field placement. It must not write the case answer, fabricate evidence/tool results, supply the reference solution/rationale/rubric for the SME, answer unrelated requests, publish rules, approve tasks, determine payment or operate real accounts. Read-only teaching examples use a different case and have no Insert answer action.

"Use another LLM to write this answer" is prohibited by the author agreement. Technical controls cannot guarantee that a person did not consult an external device/model or retype generated text. Consent, typing friction, revision history, similarity review and expert judgement are evidence/signals, not an AI-use detector or automatic misconduct decision.

Turn processing and durable memory

1. Authenticate; enforce tenant/project/task ownership, assignment, allowed editor, editability, consent and request size.

2. Inspect text for sensitive patterns and explicit instruction overrides. Treat task documents and messages as data, never higher-priority instructions.

3. Persist the authored turn with client-message ID and active revision. Enqueue one claim-once facilitation job. A replay must not create another paid attempt.

4. Build context from the template schema/help, current canonical case, relevant prior answers, unresolved issues, approved project rules, actual reviewer feedback and privacy/visibility constraints.

5. Call the configured provider outside a DB transaction. Limit output, duration and cost. Do not grant general browser/tool access.

6. Parse typed output; reject additional properties, forbidden paths and invalid references. Trace each substantive change to SME-authored spans; numeric/critical choices need explicit SME input.

7. Reconcile affected fields. Reject unsupported new facts, unsupported obligations and stale proposal context. The model can ask the SME to supply missing content; it cannot fill it itself.

8. Present proposal/diff and one task-specific follow-up. On confirm, lock/recheck revision and policy digest, apply atomically, increment revision and invalidate declarations/review where needed.

9. Update graph/readiness from confirmed data. The SME inspects/reconciles, declares the current version and submits through the normal check service.

Current implementation caveat: source-linked full-section proposals and deterministic facilitatorQuestion() are implemented, but the visible follow-up is currently selected by a limited server rule set. It is not a fully general

policy-aware adaptive planner. The new design must add a safe question-plan schema and server-rendered question templates, with model-selected allowed targets/citations, rather than directly exposing unrestricted model prose.

Proposed safe response contract

A facilitation result contains proposal ID/context revision, affected allowed fields, exact author turn/span IDs, preserved unchanged content, rule-version links, unresolved dependency IDs and a question target. The question target refers to a registered field/help/approved rule; the server renders the wording. Keep free model replies internal. Unsupported semantic conclusions are review signals, never authoritative changes. Do not log hidden reasoning; retain decision-relevant explanations and source references.

Guardrail implementation matrix

Control	Enforcement and expected behaviour	Current status / limit
Task scope	API ownership and project editor gates before reads/writes.	Local implemented; retain regression tests.
Off-topic requests	Reject or redirect to the current field using an allowed intent/question target.	Narrow generationRequest patterns/server-owned replies exist; robust multilingual intent handling is a gap.
Answer-generation requests	No substantive response; ask SME to type their reasoning.	Source-wording guard and prompt exist; not proof against external authorship.
Prompt injection	Separate untrusted policy/evidence; no tool privileges; known intent/path allowlists.	Regex tripwires exist, not universal injection protection. Add adversarial corpus and output capability gates.
Structured output	Strict schema plus additionalProperties=false and bounded arrays/text.	Existing proposal validation; each new provider needs contract qualification.
Source fidelity	Each new fact/quantity/condition maps to authored source spans.	Literal-source checks exist and can reject useful paraphrases. Improved span-based review is proposed.
Hallucinated evidence	Reject unknown IDs, unseen tool results and unsupported claims.	Known-reference checks exist; factual entailment needs human judgement.
Hallucinated policy	Only published rule IDs; correct version, source locator, applicability and exceptions.	ID/digest checks exist. A valid citation can still be misapplied.
Contradictions	Surface conflicting names, dates, identifiers and dependent findings.	KYC deterministic checks cover a subset. General cross-field dependency engine is a gap.
Adverse-media errors	Preserve allegation status, subject-match uncertainty, source date and credibility.	Template requirement; no unsupported adverse real-world assertion or automatic risk judgement.
Personal/confidential data	Type/format limits, sensitive patterns, least-data context and private storage.	Pattern checks are incomplete; no guarantee of universal PII detection.
Agent-view leakage	Server projection and visibility/reveal policies, not client hiding.	Basic allowlist exists; richer nested/per-role/graph projections need tests.
Approvals/payments	Model output cannot invoke business commands or write provenance.	Existing command boundaries retained; new stages need explicit permissions.
Authorship and paste	Consent, declarations, no insert-example action, text paste/cut/drop restriction.	Implemented friction; allow approved uploads and accessible alternatives through a reviewed exception.
Similarity/plagiarism	Compare against authorised corpus; reviewer sees matched passages and exclusions.	Existing similarity is not internet-wide plagiarism detection or proof of AI use.
Save/retry integrity	Save acknowledgements, optimistic revision, idempotency and uncertain-attempt handling.	Local save barriers implemented; provider telemetry/repair policy needs extending.

Example: missing SSN/Tax ID

SME: "The identity and address records use different family names; Tax ID and SSN are absent." The system preserves the name discrepancy and identifier absence as supplied observations. It asks jurisdiction, customer category, applicable published rule, exceptions and acceptable alternatives. It flags dependent evidence-gap/findings/rubric sections. It does not insert invented identifiers, assert that every applicant must supply both, mark high risk automatically, or create a denial recommendation.

If the author later says a tax identifier was supplied, preserve the correction history, resolve which field/document proves it and invalidate any finding or criterion relying on absence. A summary must not keep the superseded absence as current truth. When policy is missing or conflicting, use an unresolved state and ask the authorised reviewer.

Failure and safety UX

A blocked proposal displays a field-specific reason and retains the answer. Context-limit failure explicitly explains that memory was not silently truncated. Provider failure leaves canonical data intact and marks the attempt failed/uncertain as appropriate. Unsupported/unsafe output is not presented as a completed answer. Save failure prevents a tab/editor/preview transition and offers retry without clearing the draft. No "100% safe", "hallucination-free" or "AI authorship proven" badge is permitted.

07 Project policy ingestion, review and retrieval

Authority and scope

Policies belong to the project. PM/Admin uploads the initial set. SMEs may propose additional bank/customer documents, but those documents must pass authorised review before any rules become authoritative. The proposed SME policy-submission bridge is not the same as existing task-source upload. Current policy routes grant PM/Admin access; candidate attachment upload must not bypass that boundary.

Bank policy and customer policy are source categories, not a hierarchy that the model may invent. Record issuer, scope, effective dates, jurisdiction and precedence where approved. A customer document cannot silently override bank rules. Regulations/public standards can be a separately labelled source type; do not call them bank policy.

Ingestion pipeline

Stage	Stored evidence / action	Gate and user message
Upload request	Tenant/project, actor, declared category, title, issuer, version, effective dates, language and rights declaration.	Authorised upload/proposal route only.
File intake	Original bytes, hash, media signature, size and scanner result.	Refuse malware, unsupported file or oversize; never trust filename alone.
Content extraction	Text with PDF page or document section/line locators and extraction method.	Scans/OCR gaps flagged; no fabricated page citations.
Sensitive-data check	Suspected real identifiers/confidential matter and handling status.	Quarantine/restrict access; author sees a safe reason, not leaked values.
Policy qualification	Assess purpose, authority, prescriptive clauses, scope, version and relevance.	Candidate classifier result with reasons; human decision still required.
Candidate rules	Extract obligation/prohibition/permission, scope, exception, condition, effective period and exact source span.	DRAFT only; extraction does not publish a rule.
Expert rule/document review	Compare each rule with original page, resolve exceptions/conflicts, reject unrelated content.	Approve, request correction or reject with rationale.
PM/Admin publication	Freeze reviewed rule versions and publish a project rule set.	Separate authority; no chatbot-driven publication.
Retrieval/index	Index approved, authorised spans/rules with tenant/project/version/visibility metadata.	Pending/rejected/archived content excluded from authoritative context.
Task use and snapshot	Link rule versions to steps/criteria and snapshot at submission.	Policy changes invalidate stale proposals and affected review.

Current parser limit inspected: 10 MB per policy file, supported readable formats and configurable maximum pages. An image-only scan can yield NO_TEXT_FOUND. OCR is not automatically qualified because a parsing library exists. The project must configure a limit supporting its 20-page policy use case and verify page/line locators. Upload success or a clean scan does not establish authenticity.

"Is this really a policy?" review

Check issuer/owner, intended audience, scope, approval/effective metadata, prescriptive language, enforceable rules, exceptions, consistency and relevance. A slide deck, invoice, account statement, marketing leaflet, unrelated article or random document must not be promoted merely because it contains "policy". A genuine customer policy may be informal or lack a standard heading; allow an expert to explain acceptance rather than applying an opaque classifier as a final rejector.

Proposed qualification outcomes: policy candidate, supporting reference, unrelated/non-policy, unreadable, sensitive/quarantined. These are document-workflow outcomes, not extra task states. Store why and who decided. Suspected authenticity requires issuer/version clarification; an LLM cannot certify a document is genuinely issued by a bank.

Rule schema and UI

A reviewed rule needs ID/version, document version/hash, exact locator/quote, paraphrased requirement, condition/applicability, exceptions, jurisdiction, effective interval, precedence/conflict links, visibility and review decision. A step-rule link needs the step/role ID, reason and applicability judgement. A rubric-rule binding states how the requirement is assessed. Never mix a policy quote, an author's invented case fact and a reviewer inference without provenance labels.

For a 20-page document show upload -> extraction progress -> candidate rules with page links -> applicability selection. Ask the SME "Which of these approved rules applies to this step, and why?" Do not ask them to retype twenty pages. If the needed rule is not present, offer clarification or a policy proposal; never manufacture a rule from model memory.

Retrieval design

Start with the existing published project rule set for small libraries. For larger projects add a permission-filtered hybrid retrieval layer: deterministic rule/ID matches plus full-text search, optionally vectors, followed by a reviewed relevance rerank. Filter tenant, project, publication/version/effective scope and visibility BEFORE retrieval; recheck selected records before dispatch. Retrieve source snippets and their metadata, not only embeddings. Keep exceptions together with a rule when splitting text.

Document chunks are data. An embedded "ignore prior instructions" clause cannot grant tool access or change the bot's role. No arbitrary URL fetching or live bank contact is needed. Embeddings are derived project data; changing embedding models requires index versioning/rebuild and recall checks. A vector database does not itself provide memory, authenticity or hallucination prevention.

Policy-change lifecycle

When a policy version changes, identify affected tasks and links. Existing submissions retain their frozen snapshot; pending author proposals become stale if their digest changed. Expert review and PM approval must explicitly decide whether to return, revalidate or preserve an older benchmark version. Approved exported records are immutable; publish a new task revision/dataset record rather than rewriting prior benchmark evidence.

Required reviewer controls

Original document viewer with exact page/line navigation; extracted text/OCR confidence warning; candidate rule comparison; Bank/Customer/source type; effective date/scope; accept/edit/return/reject; conflict/precedence field; audit identity/timestamp; publication status; affected-task list. Expert access is assigned and audited. Until the SME bridge is implemented, PM/Admin performs intake through existing policy services and the interface explains the limitation.

08 Expert reviewer and PM screens

Review workspace

Open a frozen submission version. Header shows task/project/template, submission revision, policy snapshot, assigned Expert and approval responsibility. Authoring view used by the SME is provenance, not a separate review record. The Expert can inspect the same task regardless of Classic/Chatbot selection. Keep the horizontal graph below the evidence/assessment workspace, with nodes linked to field-specific comments and rubric/rule bindings.

Field/control	What the Expert does	Guard / help
Submitted revision and policy snapshot	Verify exactly what was submitted and which rules apply.	Read-only identity; never approve a moving draft.
Case brief and evidence	Check requested work, dates, source rights, gaps and consistency.	Agent brief must not expose the reference answer.
Original uploaded policy	Inspect actual pages and declared bank/customer authority.	Extraction/classifier output is only a proposal.
Rule/source comparison	Verify faithful requirement, scope, exceptions, effective date and locator.	A valid rule ID does not prove correct applicability.
Reference solution	Check supported steps, findings, recommendation and uncertainty.	No fabricated evidence or mandatory SSN/TIN assumption.
Tools and responses	Check schema, argument/result consistency, permissions and missing/error behaviour.	Recorded response is a fixture, not a live check receipt.
Graph nodes and routes	Inspect role work, joins, decisions, human interaction and bounded loops.	Missing runtime/export contract is an explicit blocker.
Rubric	Check observable full/partial/zero anchors, total weights, critical semantics and valid alternatives.	Grade correctness rather than identical prose.
Automated findings	Resolve blocking checks; explain accepted warnings where allowed.	Human override cannot silently waive schema/rights/export integrity.
Authorship/similarity signals	Review declaration and matched passages with context.	Never claim proven AI use or automatically reject/payhold from a score.
Field/node comment	Name the defect, expected correction and relevant evidence/rule.	Required for return; preserve exact target ID and revision.
Expert edit	Make an authorised correction with reason and before/after diff.	Audit reviewer identity; material changes invalidate downstream acceptance and require revalidation/author confirmation policy.
Return for correction	Send task back to Authoring with actionable comments.	Preserve same task; do not create a second payable task.
Complete expert review	Record review of this revision and route to PM.	Not final task approval; no immediate payment command.

Example review comment

"Reference step S3 asserts that both identifiers are mandatory. Link the published rule and establish the applicant category/jurisdiction or change this to an unresolved applicability finding. Update criterion C2 so it does not reward an unsupported requirement." This is actionable and names the dependent fields. "Bad quality" is not an adequate correction request.

PM final approval screen

Show the reviewed revision/hash, Expert decision, unresolved comments, policy document/rule approvals, final validation run/version, export readiness, coverage contribution and earning eligibility. PM may approve the exact eligible revision or return it. The UI must not say a benchmark ran or a payment was sent merely because a task was approved. Display "Approved; export adapter missing" when acceptance exists but executable readiness does not.

If Expert and PM are the same authorised person, record both responsibilities and the same actual actor; do not fabricate another reviewer. A project review route must state whether this is allowed. The demo can use the existing authorised Admin review route while the new two-stage engine is unimplemented.

Additional policy review

Review the SME document separately from the task. A task can be returned because the proposed policy is unreadable, unrelated or unsupported; the policy must not enter authoritative retrieval during the wait. Expert qualification/rule review and final publication authority remain distinct. If a task depends on the pending document, submission/approval is blocked with a linked reason rather than silently using the upload as authority.

Target stage transitions

SME submits -> Pending/Expert -> Expert completes review -> Pending/PM -> PM approves -> Approved. Either responsible stage can return to Authoring. Resubmission creates a new immutable submission version and renews required checks. Editing an approved case creates a new revision requiring a new review path; never mutate an exported accepted record.

Existing permission gap

Current TaskReview supports append-only ACCEPT/REWORK/REJECT decisions, with REVIEWER/ADMIN endpoints and checklist/comments. Plain PM final acceptance and a configurable two-stage engine are not established by that model. Add explicit stage/authority/version checks and migration tests. Preserve existing review, payment and dataset behaviour rather than renaming statuses and assuming equivalent semantics.

09 End-to-end user journeys

J1 PM creates and staffs a project

PM selects published template/version, sees fixed agent mode, chooses Classic/Chatbot/Both and the authorised review route, sets accepted target/coverage/dates, attaches policies and JD, reviews qualified SME matches and assigns work. Project summary explains exactly what is ready or missing. Existing identity/onboarding and staffing checks remain authoritative. Template field/rendering support and policy gaps must be visible before presenting a project as end-to-end ready.

J2 SME authors in Classic

SME opens an assigned task, accepts the authoring terms, reads the template Instructions PDF and fills the eight-section target form. They add repeatable evidence/tool/rubric items explicitly. Tabs retain answers; autosave acknowledges the final change before leaving. Horizontal flow appears below and reads the same confirmed data. SME selects a role to inspect its fields, checks agent visibility, resolves validations, declares current revision and submits. Current generic pages are transitional and need the registry-driven form implementation.

J3 SME authors through the common chatbot

The bot loads the actual template, saved task, approved rules and prior turns. It asks for the missing field in context. The SME describes an inconsistent name and absent identifiers. The bot preserves those observations, asks applicability/jurisdiction questions and flags affected findings/rubric; it does not write the reference decision. SME supplies the substantive content and confirms coordinated placement. The graph and Classic fields update together. No provider response means no invented update.

J4 Switch Classic and Chatbot without loss

In a Both project the SME types an unsent chat answer, switches to Classic, changes a field and immediately switches back. Save barrier completes before switching; chat draft remains. Reload restores confirmed fields and the persisted draft. If save fails, remain in the current view with error and answer intact. A pending proposal tied to an older revision is discarded/recomputed, not silently overwritten. Classic-only/Chatbot-only projects reject alternate editor writes at the API.

J5 Add a bank/customer policy

SME uploads through the proposed policy-submission route, labels authority and supplies metadata/rights. System extracts/scans/qualifies it into a pending queue. Expert checks the original text, applicability and candidate rules. Authorised publication makes only reviewed versions active. The SME binds approved rules to steps/criteria. Until that bridge exists, PM/Admin performs project-policy intake and SME task-source uploads remain non-authoritative attachments.

J6 Expert review and rework

Expert opens frozen submission, checks evidence/policy/reference/rubric and graph, reviews declarations/similarity without assuming misconduct, and records field-specific corrections. Return moves the same task to Authoring. Chat receives actual comments and current policy snapshot; Classic opens the corresponding fields. SME changes, revalidates, redeclares and resubmits a new submission version. Existing acceptance/payment counts are not incremented by rework.

J7 Expert -> PM approval

Expert completes review; task stays Review/Approval Pending, now assigned to PM. PM verifies the reviewed revision, policy approval, blockers and export readiness. Approve locks acceptance and appends the immutable dataset record; eligible earning follows existing rules. If the same person holds both roles, record both decisions honestly. This target flow requires implementation beyond current one-stage review.

J8 Graph with decisions, human review and loops

A multi-agent template provides three specialist roles and a consolidator. SME defines inputs/outputs, joins, evidence completeness, missing-evidence return, human referral, business reviewer responses, correction return and unresolved escalation. Each decision cites applicable rules and every loop has a limit/exhausted exit. Expert checks this expected process; no graph edge implies actual execution. Unsupported human-interaction/loop adapters block executable Anvil export.

J9 Completion and export

PM/Admin sees accepted count, coverage, in-progress/rework tasks and project dates. Reaching target or end date triggers a closure review, not automatic acceptance of incomplete work. Proposed end-date behaviour: stop new assignments, preserve drafts, explicitly decide rework grace and close/export eligible approved records. Freeze exact template/policy/task/adapter versions and hashes. Dataset export can succeed while executable contract export is blocked; label both statuses separately.

J10 Interruption, model outage and stale policy

A network loss preserves local text and reports unsaved state. A provider failure preserves canonical output and attempt status. The author resumes without a duplicate task/payment. A policy update invalidates stale proposals and affected pending reviews. Large context limits fail visibly; the user is not told that all prior details were remembered when they were not included.

10 Existing Praxis: reuse and explicit change flags

Audit is based on local source de758ba and prior baseline a4b4cbc. It is not a claim that these commits are deployed or that every feature is independently accepted. Do not rebuild existing modules because a diagram looks new.

Area and evidence path	Reuse / current finding	Change required
Project/template/JD: routes/projects.ts, routes/templates.ts, routes/job-descriptions.ts	Existing configuration, publishing, JD and versioning. authoringExperience local migration/choice exists.	Add versioned common authoring profiles and fixed agent-mode contract where absent; target review-route config.
Staffing: routes/staffing.ts; selection-gate.ts	Matching/assignment/identity/eligibility exist.	Reuse; automated JD preparation and Expert matching need their own qualification if added.
KYC: packages/shared/src/kyc.ts; kyc-workbench.tsx	Typed KYC fields, MCP fixtures, deterministic checks and W01-W06 forms exist.	Map to common target sections without breaking stored field names or checks.
Generic Classic: candidate/components/work/task-page.tsx	Shared title/body form exists for other workflows.	Build template-schema renderer; generic form is not full per-template anatomy.
Conversation: tasks/conversation/{service,assistant,guards,authorship}.ts	Durable history/proposals, Vertex Claude/mock, source and revision checks exist.	New MaaS adapter, richer safe adaptive questions and dependency reconciliation; no separate chatbot per template.
Autosave: use-autosave.ts; task-page.tsx; kyc-workbench.tsx	Local save barriers, metadata resume, failure retention and agent-preview preservation.	Preserve semantics in new renderer and graph; avoid remounting dirty editors.
Policies: routes/policy.ts; policy/{documents,rules,rule-sets}.ts	PM/Admin upload, document versions, extraction, rule review and publication.	SME proposal queue, reviewer access, policy-vs-random qualification, OCR/large-library retrieval acceptance.
Review: routes/task-review.ts; tasks/review-service.ts; TaskReview	Existing one-stage authorised ACCEPT/REWORK/REJECT and pinned comments.	Explicit Expert -> PM workflow, three-state UI projection and new authority checks.
Payments: pay/ledger.ts; routes/payments.ts	Eligibility/rate-card/pay-route and payout controls exist.	Do not replace. New review-stage events must not duplicate earnings or release payment early.
Dataset/export: dataset/approved-records.ts; routes/dataset.ts	Immutable accepted record/hash chain and NDJSON export exist.	Per-template executable Anvil contracts; preserve 409 NO_EXECUTABLE_CONTRACT when absent.
Workflow graph	Earlier dependency map and design previews exist.	Typed decisions, human interaction, bounded loops, horizontal layout and matching runtime/export qualification.

All paths above are relative to the inspected Praxis checkout. Source names are evidence locations, not a guarantee of interface stability. Existing TaskInstance output and project/template IDs remain canonical. The graph UI and new template profiles must not create a parallel database of answers.

Intentional local behaviour already changed

New authoring-project callers require an explicit authoring experience. NULL legacy remains previous access; no silent backfill. Classic PUT is denied for Chatbot-only; chat mutation is denied for Classic-only. New configured projects require common consent/declaration and disallow generative draft/redraft paths. Both uses a save-before-switch barrier and persisted unsent answer. KYC section changes save first; a failed save retains the current section. View as Agent saves first and retains the mounted editor. Paused projects cannot mutate chat. Annotation retains its path and read-only phone behaviour is preserved.

Changes to flag before implementation/release

1. New uniform eight-section renderer changes layout/navigation for generic templates; assess legacy saved outputs and specialised domain controls.

2. Adding graph fields changes schema, validators, agent visibility and export adapters. Version it; do not reinterpret existing dependsOn as a loop.

3. SME/Expert policy access changes permissions and introduces a publication queue. Audit tenant/project scope and rights.

4. Two-stage approval changes event timing, record immutability and payment eligibility. Prove no duplicate acceptance/earning/export.

5. Safer question planner may change the chatbot's current deterministic wording; measure supported extraction and off-topic refusal, not conversational fluency alone.

6. MaaS migration changes auth, model IDs, structured-output support, reasoning/output billing, latency and data-processing commitments. Keep provider-independent semantics and rollback.

7. Document retention, consent and typing restrictions need accessible alternatives and project-policy notice. No unannounced surveillance or secret monitoring of external devices.

Evidence boundary

Previous full local API run: 724 pass / 5 explicit gated skips, with separate later Classic-only focused lifecycle coverage. Candidate 23, console 55, shared 192, MCP replay 2 passed in the prior local delivery. The persistence follow-up added an actual Chrome/API/SQL journey across 14 catalogue templates, with final-keystroke/tab saves, failed-save retention, reload and editor switching. It also passed candidate typecheck and production build. None proves the new graph runtime, new MaaS model quality, new two-stage approval, live payouts or deployed acceptance.

11 Google Vertex Model Garden: low-cost MaaS plan

Research checked 10 November 2026. This is a selection/implementation plan, not permission to enable billing or evidence of live account access. Google's current documentation also uses the name Gemini Enterprise Agent Platform for these services. Use the exact current model card and endpoint instructions, not an old copied ID.

Recommendation

Use **gpt-oss-120b as the cost-first pilot candidate**, compare it with **Gemma 4 26B A4B IT**, and retain **DeepSeek-V4.1-Flash** as an additional challenger for cases that justify its cost. There is no measured Praxis quality winner yet. Choose the lowest cost per accepted, correctly facilitated task that passes the same safety/schema/grounding tests. Do not choose from a general coding leaderboard.

Google's current managed catalogue lists all three. MaaS is managed/serverless inference, so no dedicated GPU endpoint is required for this route. Open weights or an open model listing do not mean free hosted inference. This exercise does not require training or fine-tuning. Licence, region/data processing, quota and project availability must be checked before activation. [G1]

Pilot candidate	Input USD / million	Output USD / million	Proposed role
gpt-oss-120b	0.09	0.36	Cost-first text facilitator pilot.
Gemma 4 26B A4B IT	0.15	0.60	Google open-model challenger for the same task suite.
DeepSeek-V4.1-Flash	0.30	1.20	Challenger where reviewed task results justify it.

Prices are standard text list-price assumptions, excluding guaranteed cache discounts, taxes and negotiated terms. Google lists cheaper gpt-oss-20b pricing, but the MaaS retirement page says gpt-oss-20b-maas retired on 21 October 2026. Do not recommend an endpoint merely because it remains on a pricing page. Older Qwen3, DeepSeek V3.x/R1 and selected GLM endpoints have the same retirement issue. [G2-G4]

Google documents structured output for current managed open models. Treat that as syntax/schema capability, not evidence that the model preserves every existing field or cites a policy correctly. Always run the application validators. Implicit caching is model-dependent, Preview and not guaranteed; do not assume it for gpt-oss or deduct savings before measured usage. [G1, G5]

Pilot before selection

Use 200 reviewed, authorised conversations distributed across the actual templates, including Classic-to-chat transitions, name/identifier contradictions, policy exceptions, evidence changes, unresolved human handoffs, rubric changes and rework. Add adversarial and off-topic cases; keep training/acceptance examples separate if model work is later authorised. Use the same task/schema/policies and output contract for each provider.

Proposed go/no-go gates: zero observed cross-tenant/private-answer leaks or forbidden approval/payment mutations; zero accepted unsupported substantive answers in the safety suite; at least 99% schema-valid results without silent repair; at least 95% expert-approved clarification/placement on the reviewed set; measured latency and dollar limits acceptable to the owner. These are proposed acceptance thresholds, not measured results or proof of zero future risk. Review failures by template; aggregate scores can hide a broken domain.

Provider adapter steps

1. Verify Google Cloud project/API access, selected model card, permitted location, quota, launch stage, terms and data-processing requirements. Do not infer project availability from the public catalogue.

2. Use a backend service identity/ADC with scoped permissions. Never expose a cloud credential in SME code or ask an SME to paste an API key.

3. Implement a provider-neutral MaaS adapter behind ConversationAssistant. Translate strict response schema, allowed fields and refusal/error handling. Obtain the exact model ID/API shape from the current Google card; do not guess it from a display name.

4. Keep max output/time/cost and concurrency bounded. Existing Claude path has its own SDK contract; changing its model string is not an open-model adapter.

5. Record request/run IDs, model version, template/task/policy revision, usage input/output/reasoning/cache tokens and outcome. Avoid raw private text in telemetry.

6. Preserve claim-once jobs and human confirmation. Classify timeout/unknown completion as uncertain; no automatic duplicate paid attempt. An explicitly authorised retry is a new linked attempt.

7. Qualify with mock HTTP contracts, then an owner-authorised small live pilot. Compare expert judgement and measured cost. Roll back provider configuration without rewriting task history.

Token budget for 100,000 accepted tasks

Planning assumptions: all tasks use chat; 20 calls per original task; average 10,000 input and 1,000 billed output tokens per call; 25% allowance for rework, abandoned work and repairs. Total: 25,000 million input plus 2,500 million output = 27,500 million tokens. Count repeated context on every request, not just new words. Billed reasoning and attempted/invalid model outputs must be included according to the selected model's receipts.

At those assumptions gpt-oss-120b authoring inference is USD 3,150 total, or USD 630/month over five equally busy authoring months. Gemma is USD 5,250 total; DeepSeek is USD 10,500. These are arithmetic scenarios, not measured bills or equivalent quality. Six-month hosting averages must not be confused with five active authoring months. Classic-only work has no conversational inference; Both costs according to actual chat use, not double by definition. If only 40% of accepted tasks use this chat workload, multiply the authoring estimate by 0.4.

Extraction, OCR, embeddings, storage, app hosting, benchmark models/environments and labour are separate. Full benchmark execution across models/environments/repeats multiplies execution costs; it is not included in this chatbot estimate. Budget without cache savings. Start with a small pilot and replace these assumptions with median/p90 tokens and cost per accepted task before funding the 100,000-task scale.

Official sources

G1: Managed open models and caching: https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/maas/use-open-models G2: Partner/open-model pricing: https://cloud.google.com/model-garden/partner-models/pricing G3: Gemma pricing: https://cloud.google.com/model-garden/pricing G4: Endpoint retirement: https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/deprecations/open-models G5: Structured outputs: https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/maas/capabilities/structured-output

No current-source statement here proves source licensing suitability, account-region access or production quality. Recheck the cited pages and actual model card at activation; prices and availability can change.

12 Implementation sequence, controls and tests

Build order

Phase A: preserve existing local project/view/save functionality; publish the common template profile and field mapping. Pilot the full KYC anatomy in the reusable Classic renderer. Validate data compatibility before replacing the generic renderer. Expose unsupported template fields as a gap rather than apparent readiness.

Phase B: integrate shared Classic/Chatbot context, safe question planner, dependency reconciliation and the horizontal graph in the same canonical draft. Keep proposal confirmation, visibility and controls common. Register remaining templates through configuration and focused domain adapters; no separate chatbot forks.

Phase C: extend project-policy intake with SME proposals, qualification, assigned Expert review and authorised publication. Reuse immutable document/rule services and add permission-filtered retrieval only where needed. Do not use unreviewed policies to improve a chat demo.

Phase D: implement review-route stages/permissions and exact-version approval, then prove unchanged payment/dataset invariants. Demo can continue using the existing authorised final-review path until this passes.

Phase E: implement the Vertex MaaS adapter and run contract tests without paid calls. Obtain separate authority for account activation/live pilot, choose a model from measured results and add spend limits/telemetry. No training platform in this scope.

Phase F: qualify per-template executable Anvil mappings, especially decisions, human response and loops. Refuse unsupported contracts. Perform a real end-to-end pilot only with authorised models/environments and clearly separated acceptance evidence.

Required acceptance matrix

Test	Expected result
Each template registry profile	Required fields, enums, help/examples and checks are present in both views; unknown field types block registration.
All Classic sections	Applicable controls remain editable after first section; repeated arrays open via clear Add actions; submit section remains validation/actions.
Immediate tab/view switch	Last keystroke saved before transition. Data remains after reload; no duplicate task or extra revision from preference-only save.
Failed save / conflict	Stay in current view with text; do not claim saved; retry or reconcile exact revision.
Both/editor bypass	Both allowed; Classic-only rejects chat writes; Chatbot-only rejects Classic PUT; annotation and legacy behaviour preserved.
Chat memory	Added/corrected SSN/TIN/name facts correlate with prior evidence; superseded facts no longer treated as current.
Hallucination controls	Unknown source/rule/step IDs and invented tool results cannot apply. Correctly cited but misapplied rules go to applicability review.
Off-topic/generate request	No substantive answer; safe task-specific redirect. Multilingual, obfuscated and document-injection examples tested.
Authorship	Consent/declaration tied to exact revision; pasted text friction and legitimate upload/accessibility paths behave as specified.
Policy intake	Unrelated/malicious/unreadable/pending documents cannot become authoritative; locators and original bytes retained.
Retrieval access	No cross-tenant/project/private-rule or superseded-policy retrieval, including guessed IDs and vector reranking.
Horizontal graph	Connected decisions/human steps/joins/loops; branch labels and exhausted exits visible. Both views preserve graph/selection.
Graph semantics	Unbounded cycles, invalid handoff schemas, unknown roles and unsupported human runtime bindings block executable export.
Agent projection	Private rubric/solution/defects/discussion and other-role/future artifacts absent from initial payload and downloads.
Expert/PM route	Unauthorised role cannot act; exact revision reviewed; changed content invalidates prior decisions; same-person responsibilities auditable.
Rework/payment	Rework/resubmission retains task; one eligible earning per accepted unit; no payout from model/proposal completion.

Test	Expected result
Export	Only accepted scoped records; hash/version preserved; missing executable contract returns explicit blocker.
Model failure/retry	No lost answer, duplicate queued run or automatic paid retry of uncertain call; receipts accurately labelled.
Regression	Existing project, staffing, policy, review, payment, annotation and dataset checks pass after relevant changes.

Nonfunctional checks

Validate accessibility and keyboard-only operations, readable graph/list fallback, mobile read-only expectations, realistic concurrency, upload limits/OCR failures, recovery from interrupted writes, backups/restore, retention deletion without breaking immutable audit obligations, and explicit cost/latency monitoring. Targets require measurement; do not infer them from a successful build.

Release gates

Source review -> type/build/unit checks -> isolated SQL/API tests -> real browser journeys -> domain Expert acceptance -> authorised live-model pilot -> per-template Anvil contract qualification -> staging identity/permissions and operational evidence -> separate deployment approval. Do not conflate controlled providers with real selected models or a staged JSON representation with native execution.

Existing receipts and remaining work

Local source de758ba has save-preservation changes. Actual Chrome/API/SQL persistence check covered KYC plus thirteen other catalogue templates. Candidate component tests (23), typecheck and production build passed. Earlier broader suites are recorded in VALIDATION-REPORT.md. This documentation pass does not rerun all product tests or implement the new phases above. Graph mockups have been browser checked, but remain design previews.

13 Detailed help copy and role operating guides

Instruction PDF structure for every template

Keep six sections in each domain specification: 1 Task anatomy; 2 Synthetic/test-data rules; 3 Expected solution and rubric; 4 Author workbench; 5 Automated checks and review; 6 Example, acceptance criteria and open questions. These are specification chapters, not the same as the eight navigation sections. Each template PDF must explain what the user will produce, what each field means, what to enter, an illustrative example, validation expectations, what remains private and how corrections work. Review pages must explain the Expert/PM's actions separately.

Do not use "fictional" in SME-facing copy. Use authorised test-data guidance without implying that real personal/confidential data is automatically allowed. Instructions should explain real professional judgement and realistic case complexity. Examples teach the structure; they are not insertable current-task answers.

Exact reusable help copy

Location	Suggested text
Welcome	"Create one complete benchmark task. Supply the case, evidence, expected work and grading guidance in your own words. Your work saves to this assigned task."
Classic/Chatbot switch	"Both views use the same task. Confirmed fields and your unsent chat answer are retained when you switch."
Save in progress	"Saving your latest changes. Please wait before leaving this view."
Failed save	"Your answer is still here, but the server has not confirmed the save. Retry before switching or leaving."
Stale revision	"This task changed in another session. Review the saved version and reconcile your unsaved wording before continuing."
Empty records	"No items recorded yet. Choose Add to open the fields for an evidence record, response or criterion."
Chat role	"I can ask task-specific questions and organise your wording. You must supply the substantive case and reference answer."
Off-topic	"I can help with this assigned task. Please answer the current field or select another task section."
Generate-answer request	"Please write your own assessment. I can explain the field and identify missing information, but I cannot author the answer for you."
Missing identifier	"Which jurisdiction and customer category apply? Identify the approved rule or exception before treating this identifier as required."
Policy selection	"Select an approved rule and explain which step it applies to, including scope and exceptions."
Policy upload pending	"This document is pending review. Its rules cannot yet be used as project authority."
Non-policy upload	"This appears to be supporting material rather than a policy. Explain its purpose or submit an appropriate policy for review."
Proposal	"Review the supplied wording, its field placement and affected dependencies. Confirming updates this task revision; it does not approve the task."
Graph	"This is the process the benchmark agent should follow. It is not a record of an executed run."
Human node	"This business reviewer belongs to the case. Expert review and PM approval of your authored task are separate."
Loop	"Specify the return destination, exit condition and maximum cycles. State what happens if the issue remains unresolved."

Location	Suggested text
Agent view	"Inspect only what the benchmark agent will receive. Private reference answers and grading must not appear here."
Similarity	"These matches need human interpretation. Shared policy quotations and templates may be legitimate. A match does not prove misconduct."
Submit	"Submit this saved revision for the configured human review. Unresolved blockers must be corrected first."
Approved, not executable	"Task approved. Executable benchmark export is blocked until the required contract/adapter is supplied."

SME quick guide

Read the assignment and instruction PDF. Check template mode and project editor availability. Author the case before the reference answer; create stable evidence IDs and explicit gaps. Link only approved project rules. Supply each allowed recorded tool response and expected step yourself. Define supported findings, recommendation, uncertainty and rubric anchors. Inspect the horizontal graph, any human/loop conditions and View as Agent. Resolve cross-field changes and save warnings. Declare your own work for the current revision and submit. On rework, open pinned comments and resubmit the same task after corrections.

Expert quick guide

Check exact submission and policy versions. Read the original evidence/policy, not only the model extraction. Verify applicability, exceptions, source locators, findings, tools, graph routing, human authority, loop limits and grading alternatives. Resolve automated blockers; interpret similarity/authorship signals with judgement. Name fields/criteria/nodes in correction comments. Record edits and reasons. Complete expert review for that revision or return it. Do not imply final PM approval or a successful benchmark run.

PM/Admin quick guide

Prepare template/project/JD/policies/qualification/assignments. Review authority and pay terms before publishing work. At final review check Expert completion where required, the exact version, open blockers, policy approvals, coverage and export status. Approve or return; preserve the existing earning/payout process. Close intake deliberately at target/date and export only approved eligible records with version/hash evidence. Missing execution mappings remain blocked, not replaced by a demo result.

Help content maintenance

Version help with the template contract. Store field key, user label, definition, requiredness, conditional rule, approved example, common mistake and reviewer check. The same help registry feeds Classic tooltips, chat explanations and PDF generation. Domain owners review changes. A wording change must not silently alter an enum, schema or scoring rule. Keep a change log and regenerate affected guides after a schema revision.

14 Conversation decisions and continuation handoff

This register consolidates decisions visible in the supplied conversation and inspected source documents. It is not a verbatim transcript or a claim to remember unavailable chat history. Keep this file as the continuation index and append future dated decisions rather than relying on model memory.

Decision	Current instruction / supersession
Authoring purpose	SMEs create realistic, complete benchmark tasks with professional judgement; do not dilute the experience with "fictional" labels.
Single/multi scope	Agent mode is fixed by the published template. Independent repeated cases are single-agent trials.
Shared chatbot	One chatbot engine for every template; template-driven context/questions/checks. No per-template chatbot forks.
Shared Classic	Common schema-driven framework with domain configuration/plug-ins. Reuse common screens instead of fourteen independent implementations.
Project experience	PM/Admin chooses Classic, Chatbot or Both during project creation. Both shares one task and retains draft answers.
Latest layout	Authoring fields or conversation on top; horizontal connected graph BELOW in both views. This replaces side preview and the tall vertical graph as the desktop target.
Graph semantics	Real connected nodes, decision diamonds, explicit branches, business human review and bounded loops. Text-card chains are insufficient.
Human distinction	A business human interaction can be an authored graph node. Expert/PM platform task review remains a status outside it.
Navigation	Show section and step counts separately. Every applicable Classic content section has its fields/Add actions. Save on transition; never lose the answer.
Agent preview	Available in both editors, with server-enforced private-answer/rubric/role visibility.
Task states	Authoring / Review-Approval Pending / Approved, with responsible stage inside Pending. Rework returns to Authoring.
Review routes	Implement SME -> PM and SME -> Expert -> PM. Demo uses first route through currently authorised final reviewer. Expert and PM may be same authorised person with honest decision records.
Excluded comparisons	Two-SME winner selection and SME versus LLM authored-task competition excluded from MVP; enterprise exploration deferred.
Policy scope	Base documents uploaded at project level. SME bank/customer additions require qualification/review before authority.

Decision	Current instruction / supersession
Document authenticity	Detect unreadable/non-policy/unrelated content and require expert judgement. A classifier or filename does not prove policy authenticity.
Rule applicability	Extract candidate rules with citations, then ask which approved rule applies to each step, including exceptions.
Chat memory	Correlate new answers with saved fields/history/policies. SSN/TIN absence is an observation, not an invented mandatory requirement.
Human authorship	Bot facilitates, extracts and clarifies; SME supplies answers/reference/rubric. No Generate my answer button.
External LLM use	Prohibit through consent/declaration and review controls, but do not promise technical proof or perfect prevention.
Typing and similarity	Authoring paste/cut/text-drop friction plus legitimate document upload and reviewed accessibility exceptions. Similarity is a human review signal, not automatic misconduct/payout determination.
Instructions	Examples for every section and template PDFs: what to expect, field meanings, what to do and how to review.
Existing Praxis	Inspect/reuse existing services. Flag changes, missing permissions and unqualified adapters; do not rebuild everything or claim completion from a mockup.
Validation	End-to-end UI/API/data checks and regression impact required. Controlled providers are not live-model acceptance.
Payments	Accepted task eligibility follows existing terms/ledger. Rework/resubmission is not a second payable unit.
Completion/export	Target accepted count/coverage or end date prompts closure/export decision. Only approved eligible versions export; executable Anvil contract qualification is separate.
Models	Cost-first managed open-model inference on Google Vertex Model Garden/MaaS. "Open source" was clarified to mean free/cheap usage; hosted calls still cost money.
Training	No training required for this authoring exercise. A future model-training platform is a separate discussion/scope, not silently included.
Scale/cost	Planning scenario 100,000 tasks over five authoring months; six-month host budget separates development and inference/benchmark costs. No quoted estimate is a measured bill or spend authority.
Handover	Include requirements, field guides, journeys, source gap audit, chatbot/guardrails, policy/RAG, model plan, reviewer/help guides and all relevant decisions in Praxis handover.

Decision	Current instruction / supersession
Figma	Latest request is Praxis handover, not another Figma pack. Preserve prior files but do not use Figma delivery as a substitute for the specification.

What exists locally

Praxis checkout: /Users/vasudevansundarababu/Documents/Codex/praxis-conversational-authoring. Clean local product commit de758ba contains the persistence fix; 09aa194 contains project experiences/common controls and 38bbaa3 earlier conversational work. None is asserted pushed/deployed by this handover. No new paid model call is part of it.

Local graph reference: praxis-classic-chatbot-flow-preview.html in the thread visualization directory, now horizontal at desktop widths. It is an interactive design preview with confirmation/retained text, not a live LLM or executable multi-agent runtime. The handover package includes standalone preview and Classic/Chatbot PNG references.

What to build next

Follow chapter 12 phases. Start with template registry/full Classic profile and safe shared question planner. Keep output schema/version and review/payment invariants. Add policy proposal bridge and two-stage approval explicitly. Qualify MaaS adapter and per-template Anvil contracts only with authorised live acceptance. A graph mockup, dataset export or SDK mock response must not be labelled full production acceptance.

Open decisions, not invented answers

Owner/domain reviewers must confirm all published template agent modes, final field/outcome contracts outside KYC, review-edit authority and author reconfirmation rules, same-person approval policy, deadline/rework grace, per-project document rights/retention, minimum mobile editing support, operator budget caps and live-provider pilot acceptance. These do not prevent delivery of this specification; they do prevent unsupported rollout claims.

Next-session checklist

Read this index, then 00 and 10. Confirm current branch/status before editing. Inspect current sources instead of replaying stale older instructions. Update the decision register and affected chapters for each material correction. Record implementation commit, checks actually run, failures/limits and deployment status. Avoid secret/config/runtime credential reads. Keep Anvil/Praxis independence and cloud-spend authority separate. Do not restart old agents or treat monitoring prompts as coding instructions.

15 KYC exact field contract and checks

Source: packages/shared/src/kyc.ts, packages/shared/src/mcp.ts and current workbench. These are existing canonical names; labels can be friendlier, but adapters must not silently rename them. The earlier draft's parties[]/documents[] and risk_rating are not literal top-level current KycTask properties. Current paths are customerFile.parties[], evidence.documents[] and solution.riskRating. Anvil reference_output is an export mapping, not a license to replace current solution storage.

Fields are optional in the TypeScript draft interface so incomplete work can save. Submission requiredness is enforced by kycChecks/slot rules; the interface's optional marker does not make a complete benchmark field optional. The target template profile must encode those checks rather than invent different requiredness.

Exact current field(s)	Meaning / author action	Important check
header.title	Case title.	Required useful assignment name.
header.difficulty / estimatedSteps	Complexity and expected work count.	Difficulty 1-5 in workbench; no execution claim.
header.jurisdiction / language	Case boundaries.	Supplied scope; not inferred from identifier name.
header.coverageTags / edgeCase	Slot classification.	Configured coverage dimensions and assigned edge case.
scenario.trigger	Trigger.	NEW_ONBOARDING or PERIODIC_REFRESH.
scenario.customerType / product / channel	Customer/service/context category.	Align with evidence and policy applicability.
scenario.businessContext	Agent-visible brief.	No private solution; no invented case fact.
customerFile.parties[].id	Stable party ID.	Unique; referenced documents/ownership must resolve.
parties[].role / kind / name	Role, PERSON/ENTITY and authorised test identity.	Use catalogue role; do not insert real sensitive identity.
parties[].dateOfBirth / incorporationDate	Relevant supplied date.	Valid date; correct kind.
parties[].nationality / incorporationCountry	Supplied jurisdictional attributes.	Do not assume tax/identity requirements solely from this field.
parties[].addresses	Supplied addresses.	Authorised test data; record uncertainty.
parties[].identifiers[].type/value	Supplied identifier and value.	Missing stays missing; use approved synthetic formats/permissions.
customerFile.ownership[].ownerId / ownedId / percent	Ownership link.	Valid parties and meaningful quantities; not unsupported control inference.

Exact current field(s)	Meaning / author action	Important check
customerFile.statedPurpose / expectedActivity	Customer's stated business facts.	Distinguish statement from independently verified evidence.
evidence.documents[].id/type/issuer	Record ID, document kind and issuer.	Unique ID; missing issuer stays unknown.
documents[].partyId	Party concerned.	Existing party ID.
documents[].issueDate / expiryDate	Validity information.	Valid dates; inconsistency/expiry reflected in solution.
documents[].fields[].name/value	Extracted record facts.	Faithful authorised source content.
documents[].defects	Private evaluator-facing defects.	Exclude from initial agent-visible projection.
environment.tools[]	Allowed catalog tool keys.	No unknown action or operational permission.
environment.responses[].id/tool	Recorded fixture and tool.	Stable response ID; declared allowed tool.
responses[].input / arguments	Summary and MCP argument object.	Arguments satisfy tool schema; legacy input handling preserves compatibility.
responses[].kind / response	Recorded result.	CLEAR/HIT/NEAR_MATCH/DATA/ERROR; do not claim live execution.
policy.rules[].id/text	Existing task policy free-text representation.	Chat cannot rewrite project authority through this section. Published rule links use policy service metadata.
solution.steps[].id/action	Expected step, TOOL or REASONING.	Unique ID; current type does not encode arbitrary decisions/humans/loops.
steps[].tool/responseId/input/arguments	Expected recorded call binding.	Must resolve to matching fixture/tool arguments.
steps[].expectedResult / why / dependsOn	Expected output, rationale and prerequisites.	Known dependencies, no self-reference; new graph loop contract is separate.
solution.findings[].id/description/evidenceRefs/severity	Supported conclusion.	Known evidence references; LOW/MEDIUM/HIGH severity.
solution.disposition	Recommended next action.	RECOMMEND_APPROVE, RECOMMEND_REJECT, REQUEST_INFORMATION or ESCALATE.
solution.riskRating / riskFactors	Supported risk classification and reasons.	LOW/MEDIUM/HIGH; no automatic real-world risk decision.

Exact current field(s)	Meaning / author action	Important check
solution.requestsToCustomer	Expected information requests.	Specific gaps, permitted action boundaries.
solution.rationale / humanHandoff	Evidence-grounded explanation and authorised handoff.	Existing human-gate checks for approve/reject/high risk; avoid autonomous finalisation.
rubric.criteria[].id/name/weight/passCondition/critical	Observable criterion and score contribution.	Total 100; references and critical semantics require review.
rubric.commonErrors[].description/caughtBy	Plausible wrong path and catching criterion.	Valid criterion reference.
rubric.acceptableVariations[]	Other valid approaches.	Do not force identical wording or unsupported exclusions.

Allowed tools and pick lists

Tools: document_check, registry_lookup, sanctions_pep_screening, adverse_media_search, address_check, risk_calculator, request_info. Tool argument schemas live in the shared MCP registry; do not hard-code a new call schema in the chatbot.

Party roles: CUSTOMER, DIRECTOR, SHAREHOLDER, BENEFICIAL_OWNER, TRUSTEE, SETTLOR, BENEFICIARY, SIGNATORY, OTHER. Response kinds and dispositions are listed above. Free-form labels do not extend a published enum.

Richer rubric and graph gaps

Current criteria do not directly contain every target field such as partialCredit, zeroCredit, ruleVersionIds or per-role bindings. Current KycStep does not declare actor permissions, human response schemas, branch conditions or bounded loops. Add these through a reviewed schema/version and adapters; never claim that a visual diamond is already a stored executable condition.

Export mapping

Export adapters explicitly map existing scenario/customer/evidence fields to agent-visible evidence, applicable policy/tool/action boundaries to constraints/actions, ordered supported steps to stages, private solution to reference_output, and criteria to rubric bindings. Exact rights, task IDs, versions and hashes accompany accepted records. If a field/graph construct cannot be represented, executable export fails with a specific diagnostic. Existing dataset export and MCP replay are separate from that qualification.

16 Template coverage and common screen generation

The local instruction catalogue contains fourteen templates. This is the coverage set for reusable authoring UI tests, not proof that fourteen fully typed/executable workflows already exist. KYC is the concrete typed pilot; most other current forms remain generic. Multi-agent KYB is a design reference; all actual published agent modes must be confirmed in template configuration.

Each template spec retains six chapters: Task anatomy; Synthetic/test-data rules; Expected solution and rubric; Author workbench; Automated checks and review; Example/acceptance/open questions. The editable domain guides in template-instructions are included as appendices. Their labels/outcome examples must be reconciled with approved schema before implementation.

Required reusable profile per template

Define exact field names/types/labels, requiredness and dependencies; pick-list catalogues; authorised test-data restrictions; tool schemas/fixtures or none; reference shape; rubric/critical semantics; deterministic checks; human-review wording; environment capability; Anvil adapter mapping; section examples and field help. Register it once. Classic renders controls; Chatbot reads the same profile and proposes only allowed field updates. Neither LLM-generated arbitrary UI nor a generic free-text form is a substitute for this profile.

Catalogue listing

DM01 - Q4 Global Paid Search Query Clustering

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: query dataset, product brief, locale and channel constraints. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: clusters, query assignments, intent labels, exclusions, ambiguous queries, rationale. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: intent: informational, commercial, transactional, navigational, unclear; assignment: assigned, excluded, needs_review. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

DM02 - Omnichannel Creative and Locale Compliance

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: creative assets, substantiated claims, channel specifications, locale requirements. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: asset-by-locale findings, claim evidence, proposed corrections, release recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: finding: compliant, noncompliant, insufficient_evidence, not_applicable; recommendation: ready_for_review, revise, hold. If no

choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK03 - Adverse Media Review

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: customer identifiers, supplied media extracts, dates and source metadata. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: identity match, source reliability, allegations versus established facts, relevance, recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: match: confirmed, possible, false_positive, unresolved; recommendation: close, escalate, request_information. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK04 - Audit and Evidence Preparation

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: audit request, control descriptions, evidence inventory and review period. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: evidence index, request mapping, provenance, gaps, preparation memo. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: evidence: available, missing, incomplete, conflicting, out_of_scope. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK05 - Communications Surveillance

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: synthetic communications, participant roles, timestamps and surveillance policy. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: relevant excerpts, contextual findings, policy mapping, escalation recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: disposition: close, escalate, request_context; finding: supported, unsupported, inconclusive. If no choice fits, flag it for

review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK06 - Compliance Monitoring

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: defined monitoring scope, records, obligations and monitoring period. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: obligation-to-evidence assessment, exceptions, severity, proposed follow-up. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: result: compliant, noncompliant, insufficient_evidence, not_applicable; severity: low, medium, high, critical, undetermined. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK09 - Customer Risk Rating

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: customer profile, risk factors, evidence and versioned rating methodology. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: factor assessments, calculation where supported, proposed risk, overrides, rationale. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: risk: low, medium, high, undetermined; override: none, proposed_increase, proposed_decrease. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK10 - Enhanced Due Diligence

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: high-risk case, source-of-funds/wealth evidence, ownership and adverse findings. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: enhanced checks, evidence gaps, risk assessment, recommendation, human handoff. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: recommendation: proceed_for_approval, request_information, escalate, recommend_decline; evidence: verified, unverified, conflicting, missing. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK12 - Issue and Remediation Tracking

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: issue records, action plans, owner records, due dates and closure evidence. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: issue status, linked evidence, overdue assessment, closure recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: status: open, in_progress, pending_validation, closed, reopened; recommendation: retain_open, request_evidence, recommend_closure. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK13 - KYB and Beneficial Ownership

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: entity records, ownership graph, ownership percentages, control evidence and applicable policy. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: entity verification, ownership traversal, beneficial owners and controllers, gaps, recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: party_role: legal_entity, shareholder, beneficial_owner, controller, authorised_signatory, trustee, settlor, beneficiary, other; disposition: complete_for_review, request_information, escalate. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK14 - KYC Onboarding and Refresh

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: customer case, identity evidence, prior KYC and applicable dated policy. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: steps, findings, disposition, risk_rating, rationale, evidence citations, human_handoff. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the existing KYC canonical pick lists: scenario.trigger is NEW_ONBOARDING or PERIODIC_REFRESH; solution.disposition is RECOMMEND_APPROVE, RECOMMEND_REJECT, REQUEST_INFORMATION or ESCALATE; solution.riskRating is LOW, MEDIUM or HIGH. Party roles and response kinds follow packages/shared/src/kyc.ts. These canonical values supersede provisional wording in earlier guide examples; do not silently add enum values.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK15 - AML Transaction-Monitoring Alerts

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: alert, synthetic transactions, customer profile, scenario parameters and historical context. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: transaction analysis, expected versus observed activity, gaps, alert disposition, escalation rationale. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: disposition: close_with_rationale, request_information, escalate_for_investigation; suspicion: not_established, potential, unresolved. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

BK11 - Fraud-Risk and Compliance Handoffs

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: fraud alert, transaction records, case timeline, roles and escalation policy. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: facts, fraud indicators, compliance concerns, handoff package, recommendation and rationale. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: handoff: fraud, compliance, joint_review; recommendation: request_information, escalate, close_with_rationale. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

DM03 - Noise Cancelling Headphones Launch

Common UI: Classic schema renderer + shared Chatbot + instructions + horizontal flow + agent projection + common checks/review.

2. Case inputs and evidence Input means the material the agent can inspect: product brief, test evidence, audience, channel requirements, budget and launch dates. Give each item a stable ID, source and relevant date. Record gaps explicitly. Use authorised test data; exclude real personal identifiers and confidential material unless expressly authorised.

5. Expected work and result Steps describe the work the agent should do; findings state supported conclusions. The expected deliverable for this template is: launch plan, channel assets, substantiated claims, budget allocation, measurement plan and release recommendation. Explain the rationale and unresolved issues. Keep reference answers separate from agent-visible inputs.

6. Allowed outcomes and values Use the template pick lists rather than new labels. Current proposed labels: claim_status: supported, unsupported, needs_review; release: ready_for_review, revise, hold. If no choice fits, flag it for review; do not silently invent an enum or policy threshold.

Implementation gate: domain owner confirms exact schema/pick lists and published agent mode; developer supplies tests and executable adapter capability. Do not mark full Anvil readiness from this help text.

Registration acceptance

Each profile renders all applicable fields, shows examples/help, rejects unsupported enums, preserves data during editor switches, scopes approved rules, projects agent visibility and runs its registered checks. The shared bot must not force KYC-specific names into unrelated templates. A template with only a generic pass-through validator is explicitly unqualified for domain completeness until its checks and reference/rubric bindings exist.